$x[n] = A + w[n]$, $w \sim N(0, \sigma^2 I)$

estimate $\hat{A}$, $\hat{A}_{MVU} = \bar{x}$

$\bar{x} \sim N(A, \frac{\sigma^2}{N})$

1) Find $\hat{A}_{MVU}$ by RBLS . $\check{\theta} = x[1]$

unbiased $\check{A} = x[1]$.

find sufficient statistics $T(x) = \sum x_n$

since $p(x;A) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(\sum x_n^2 - 2A\underbrace{\sum x_n}_{T(x)} + NA^2)}$

then $\hat{A}_{MVU} = E(\check{A} | T(x)) = E(x[1] | \sum x_n)$

$$\begin{bmatrix} x[1] \\ \sum x[n] \end{bmatrix} \sim N\left(\begin{bmatrix} A \\ NA \end{bmatrix}, \begin{bmatrix} \sigma^2 & \sigma^2 \\ \sigma^2 & N\sigma^2 \end{bmatrix} \right)$$

$$\hat{A}_{MVU} = E(x[1] | \sum x_n) = A + \frac{\sigma^2}{N\sigma^2}(\sum x[n] - NA) = \bar{x}$$

estimate $\hat{\sigma}^2$, $\hat{\sigma}^2_{MVU} = \frac{1}{N}\sum_{n=1}^{N}(x[n] - A)^2$

estimate $\hat{A}, \hat{\sigma}^2$ $\hat{A}_{MVU} = \bar{x}$, $\hat{\sigma}^2_{MVU} = S^2 = \frac{1}{N-1}\sum_{n=1}^{N}(x[n] - \bar{x})^2$

$$Var(\hat{\theta}) \geq \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix}$$ $\bar{x}$, S^2 independent

1) Prove $\hat{A}_{MVU}$, $\hat{\sigma}^2_{MVU}$ by NF Fraction

$p(x;\theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(\sum x_n^2 - 2A\sum x_n + NA^2)}$

$T_1 = \sum x_n$. $T_2 = \sum x_n^2$

$E(T_1) = NA$, $E(T_2) = NA^2 + N\sigma^2$, $E(T_1^2) = N^2A^2 + N\sigma^2$

So $\tilde{T}_1 = \frac{1}{N}T_1$, $\tilde{T}_2 = \frac{1}{N-1}(T_2 - \frac{1}{N}T_1^2)$ satisfy $E(\tilde{T}) = (A, \sigma^2)$

$\hat{A}_{MVU} = \tilde{T}_1 = \bar{x}$, $\hat{\sigma}^2_{MVU} = \frac{1}{N-1}[\sum x_n^2 - \frac{1}{N}(\sum x_n)^2] = \frac{1}{N-1}(\sum x_n^2 - N\bar{x}^2)$

2) Prove S^2 unbiased

$E(S^2) = \frac{1}{N-1}E[\underbrace{\sum_{n=1}^{N}(x_n^2 - 2x_n\bar{x} + \bar{x}^2)}_{\sum_{n=1}^{N} x_n^2 - 2N\bar{x}^2 + N\bar{x}^2}] = \frac{1}{N-1}[\sum_{n=1}^{N}E(x_n^2) - NE(\bar{x}^2)] = \frac{1}{N-1}(N\sigma^2 - \sigma^2)$

$x_n \sim N(0, \sigma^2)$, $\bar{x} \sim N(0, \frac{\sigma^2}{N})$

estimate $\hat{SNR}$, $SNR = \frac{A^2}{\sigma^2}$

$$Var(\hat{SNR}) = \left(\frac{2A}{\sigma^2} \quad -\frac{A^2}{\sigma^4}\right) \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix} \begin{bmatrix} \frac{2A}{\sigma^2} \\ -\frac{A^2}{\sigma^4} \end{bmatrix} = \frac{2}{N} SNR(SNR + 2)$$

$x[n] = H\theta + w[n]$, $w \sim N(0, \sigma^2 I)$

$\hat{\theta}_{MVU} = (H^T H)^{-1} H^T x \qquad Var(\hat{\theta}) \geq \sigma^2 (H^T H)^{-1}$

Pf: $p(\vec{x};\theta) = \frac{1}{(2\pi\sigma^2)^{N/2}} e^{-\frac{1}{2\sigma^2}(x-H\theta)^T(x-H\theta)}$

$\ln P = -\frac{N}{2}\ln 2\pi\sigma^2 - \frac{1}{2\sigma^2}\underbrace{(x-H\theta)^T(x-H\theta)}_{x^T x + \theta^T H^T H\theta - 2x^T H\theta}$

$\frac{\partial}{\partial\theta}\ln P = \frac{1}{\sigma^2}(H^T x - H^T H\theta) = \underbrace{\frac{1}{\sigma^2}H^T H}_{I(\theta)}\left[\underbrace{(H^T H)^{-1} H^T x}_{MVUE} - \theta\right]$

$x[n] = H\theta + w[n]$, $w \sim N(0, C)$

$\hat{\theta}_{MVU} = (H^T C^{-1} H)^{-1} H^T C^{-1} x \qquad Var(\hat{\theta}) \geq (H^T C^{-1} H)^{-1}$

$x[n] = A + w[n]$, $w \sim N(0, C)$

estimate $\hat{A}_{MVU}$ and CRLB

Linear $x = A 1_N + w$

$\hat{A}_{MVU} = \frac{1^T C^{-1} x}{1^T C^{-1} 1}$, $Var(\hat{A}_{MVU}) = \frac{1}{1^T C^{-1} 1}$ CRLB

$x[n] = A\cos(2\pi f n + \phi) + w[n]$

1) Find CRLB for $\hat{f}$

$x[n] = s[n;\theta] + w[n] \qquad Var(\hat{\theta}) \geq \frac{\sigma^2}{\sum (s')^2}$

$Var(\hat{f}) \geq \sigma^2 / 4\pi^2 A \sum n^2 \sin^2(2\pi f n + \phi)$

2) Sufficient Statistics for $\hat{f}$ doesn't exist

$p(x;f) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}\sum x_n - 2\underbrace{\sum x_n \sin 2\pi f n}_{\text{couple}} + \sum s_n^2}$

3) Find $\hat{A}_{MVU}$

Linear Gaussian, $\hat{A}_{MVU} = (H^T H)^{-1} H^T x = \frac{\sum x_n s_n}{\sum s_n^2}$

$x \sim N(0, C) \qquad I(\theta) = \frac{1}{2} tr\left[\left(C^{-1}(\theta) \cdot \frac{\partial C}{\partial \theta}\right)^2\right]$

$x[n] = A + w[n]$, $A \sim N(0, \sigma_A^2)$, $w \sim N(0, \sigma^2 I)$

estimate $\hat{\sigma}_A^2 \qquad x = A 1_N + w$, $x \sim N(0, C)$.

$C = \sigma_A^2 1 1^T + \sigma^2$, $C^{-1} = \frac{1}{\sigma^2}\left(I - \frac{\sigma^2}{N\sigma_A^2 + \sigma^2} 1 1^T\right)$ Pf: inverse lemma

$Var(\hat{\sigma}_A^2) \geq 2\left(\sigma_A^2 + \frac{\sigma^2}{N}\right)^2$

$X[n] \sim U(-\theta, \theta)$

1) Find Sufficient Statistics $T(x)$

$p(x) = (\frac{1}{2\theta})^N u(\theta - \max|X[n]|)$ $\quad T(x) = \max|X[n]|$

2) Prove CRLB cannot be achieved $E(\frac{\partial}{\partial\theta} \ln P(x;\theta)) \neq 0$

CRLB $Var(\hat{\theta}) \geq I^{-1}(\theta)$ $\quad$ Fisher Information $I(\theta) = -E[\frac{\partial^2}{\partial\theta^2} \ln P]$

If Regularity Condition $E[\frac{\partial}{\partial\theta} \ln P] = 0$

1) MVUE (Min Variance Unbiased)

$Var(\hat{\theta}) = I^{-1}(\theta)$ 称 efficient. MVU $\underset{\not\Rightarrow}{\Leftarrow}$ efficient $\hat{\theta}_{MVU}$ 不一定 CRLB

$\frac{\partial}{\partial\theta} \ln P = I(\theta)[g(x) - \theta]$ 则 $\hat{\theta}_{MVU} = g(x)$ CRLB 可达到

2) Vector

CRLB $Var(\hat{\theta}) \geq I^{-1}(\theta)$ $\quad$ Fisher Matrix $I_{ij}(\theta) = -E[\frac{\partial^2}{\partial\theta_i \partial\theta_j} \ln P]$

If $I(\theta)$ not invertible, CRLB $\to \infty$

3) Function of Parameter

If $\alpha = \varphi(\theta)$, $Var(\hat{\alpha}) \geq \varphi'^2 \cdot I^{-1}(\theta)$

Vector $Var(\hat{\alpha}) \geq (\varphi')^T I^{-1}(\theta)(\varphi')$

4) CRLB Proof

有偏时 $\frac{\partial}{\partial\theta} E(\hat{\theta}) = E(\hat{\theta} \cdot S(x;\theta))$ $\quad$ 若无偏, 则 $\frac{\partial}{\partial\theta} E(\hat{\theta}) = 1$

因为 $\frac{\partial}{\partial\theta} \int g(x) f(x|\theta) d\theta = \int g(x) \frac{\partial}{\partial\theta} f(x|\theta) d\theta$ 积分微分换序

$= \int g(x) f(x|\theta) \frac{\partial}{\partial\theta} \ln f(x|\theta) d\theta$ 技巧: $f'(x) = f(x) \frac{d}{dx} \ln f(x)$ $= E(\hat{\theta} \cdot S(x;\theta))$

$E(\hat{\theta} \cdot S(x;\theta)) = cov(\hat{\theta}, S(x;\theta))$ $\quad$ $E[S(x;\theta^*)] = 0$ 可以.

Cauchy-Schwatz 不等式 $[\frac{\partial}{\partial\theta} E(\hat{\theta})]^2 \leq Var(\hat{\theta}) Var(S(x;\theta)) = Var(\hat{\theta}) E[S(x;\theta)^2]$

故 $Var(\hat{\theta}) \geq [\frac{\partial}{\partial\theta} E(\hat{\theta})]^2 \cdot E[S(x;\theta^*)^2]^{-1}$ $\quad$ 无偏 $Var(\hat{\theta}) \geq E[S(x;\theta^*)^2]^{-1}$

Find MVUE :

1. Gaussian Linear $\hat{\theta}_{MVU} = (H^T C^{-1} H)^{-1} H^T C^{-1} x$

2. NF Fraction, $T(x)$, unbiased $\hat{\theta}_{MVU} = g(T(x))$

3. $\frac{\partial}{\partial \theta} \ln p = I(\theta)[g(x) - \theta]$ $\hat{\theta}_{MVU} = g(x)$

4. RBLS. Give unbiased $\check{\theta}$, $\hat{\theta}_{MVU} = E(\check{\theta} \mid T(x))$

完备性证明 $T(x) = \sum_{n=0}^{N-1} x[n]$

$g(T) = \frac{1}{N} T(x)$ 是唯一对 A 的无偏估计函数

假设另有 $g'(T)$，使 $E[g'(T)] = A$，则

$$E[g(T) - g'(T)] = \int [g(T) - g'(T)]\, p(T)\, dT$$

因为 $T \sim N(NA, N\sigma^2)$，故 $p(T)$ 是高斯分布 故 $g(T) - g'(T) = 0$ a.e.